# Kiron Sukesan

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## Summary

Senior software and machine learning engineer with 9+ years of experience and 2 years of management experience. Track record building large-scale production systems at Amazon and ML platforms at Compass. Research background in Bayesian probabilistic modeling.

## Skills

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| **Languages / Tools** | Java, Python, C++, Rust, JavaScript, MATLAB, Agentic Development |
| **ML / AI** | ANN, HNSW, Computer Vision, Gaussian Processes, Bayesian Inference, Semi-Supervised Learning, Deep Learning, Reinforcement Learning |
| **Storage Systems** | Distributed Database Internals, Consensus Protocols (Paxos, Raft), Database Engine Implementation (LSM tree and B-tree) |

## Experience

**Lead Engineer** *at* **Freelance** Apr 2024 – Current
*Remote*

- Architected and developed AppMan, a full-stack apartment management platform using Angular, PostgreSQL, and Docker, modernizing legacy Oracle APEX workflows for property operations, payments, leasing, taxes, and task management. In production use for 1000+ unit apartment management company.
- Designed and build a test-driven engineering infrastructure with unit, integration, adversarial, migration, authorization, and smoke test. This foundation allowed the ultize a custom agentic development workflow to accelerate delivery while maintaining human oversight and production-quality standards.

**Software Development Manager** *at* **Amazon Web Services (AWS)** Aug 2024 – Apr 2026
*DynamoDB Storage Node, Seattle, WA*

- Leader in the engineering organization for DynamoDB's Storage Node, one of AWS's highest-scale distributed storage systems, overseeing architecture, delivery, and team health. Specifically responsible for the TTL system, replication protocol, general correctness of DynamoDB Storage, and feature expansion efforts within storage.
- Responsible for hiring, mentoring, and performance management for a team of 10+ junior, mid-level and senior software engineers; established engineering standards and oncall culture.

**Senior Software Engineer** *at* **Amazon Web Services (AWS)** Jan 2023 – Aug 2024
*DynamoDB Storage Node API Team, Seattle, WA*

- Designed and implemented features for the DynamoDB Storage Node API layer, contributing to a system serving millions of customers with stringent latency, availability, and durability requirements.
- Led oncall and operational excellence initiatives, driving root cause analyses and systemic improvements to reduce customer-impacting events across the storage tier.

**Machine Learning Engineer II / Senior Machine Learning Engineer** *at* **Compass** Dec 2019 – Nov 2022
*Seattle, WA*

- Built and owned Computer Vision based ML models used by agents across the platform, directly influencing transaction outcomes.
- Architected end-to-end ML pipelines spanning data ingestion, feature engineering, model training, evaluation, and production deployment on cloud infrastructure.
- Mentored junior ML engineers and drove platform improvements that reduced model iteration cycle time and increased experiment throughput.

**Software Development Engineer I/II** *at* **Amazon Kindle** Oct 2017 – Nov 2019
*Greater Seattle Area*

- Designed and shipped features for Amazon Kindle's internal data processing pipeline, collaborating with senior engineers on distributed systems design, API contracts, and service reliability.
- Participated in oncall rotations and drove root cause analyses for production incidents, contributing to improved monitoring and alerting infrastructure.
- Promoted to SDE II after demonstrating technical ability with end to end solution for Kindle compliance with GDPR's requirement to return data to customers on request.

**Data Scientist / Research Engineer** *at* **Washington University in St. Louis** May 2016 – Aug 2017
*Laboratory of Sensory Neuroscience and Neuroengineering, St. Louis, MO*

- Developed a Bayesian active learning framework using Gaussian Process Classification to efficiently estimate audiometric threshold surfaces, reducing clinical measurement burden while preserving diagnostic accuracy.
- Designed a novel ML-based hardware calibration algorithm for audiometric equipment, improving signal fidelity and measurement consistency across experimental sessions.
- Co-authored three peer-reviewed publications on Bayesian active learning, psychometric field estimation, and online machine-learning audiometry.

**Software Developer** *at* **Purnatva Solutions Pvt. Ltd.** Sep 2014 – Aug 2015
*Bengaluru, India*

- Implemented real-time GPS bus tracking module for the Falcon Mobile App, supporting large-scale deployment with 10,000+ active users across the platform.
- Contributed to *vidyadhan.org*, a LAMP-stack scholarship portal connecting sponsors to underprivileged students, working directly with the product team on feature delivery and backend development.

**Biomedical Engineer, Undergraduate Research** *at* **Wash. U. in St. Louis** Aug 2013 – May 2014
*Department of Biomedical Engineering*

- Designed and built a wearable EEG recording device for overnight clinical use, using custom analog circuitry, embedded software, and mechanical design in close collaboration with two biomedical engineering peers and clinical partners.
- Validated device performance against clinical EEG standards and iterated on hardware and firmware based on clinician feedback.

**Neurosurgery Research Intern** *at* **University of Washington** Jun 2013 – Aug 2013
*Brain-Computer Interface Lab (Dr. Jeff Ojemann), Seattle, WA*

- Developed a signal analysis algorithm for electrocorticographic (ECoG) neural signals; diagnosed and resolved electromagnetic noise artifacts that had been corrupting recorded data.
- Constructed a closed-loop BCI device that enabled motor-impaired patient hand movement through real-time classification of theta wave activity from ECoG recordings.

## Education

**Washington University in St. Louis** 2015 – 2016
*Master of Science, Computer Science* St. Louis, MO

**Washington University in St. Louis** 2010 – 2014
*Bachelor of Science, Biomedical Engineering* St. Louis, MO

## Publications


- Song, X. D., **Sukesan, K. A.**, & Barbour, D. L. "Bayesian active probabilistic classification for psychometric field estimation." *Attention, Perception, & Psychophysics*, 80 (2018), 798–812. DOI: 10.3758/s13414-017-1460-0.
- Barbour, D. L., DiLorenzo, J. C., **Sukesan, K. A.**, Song, X. D., Chen, J. Y., Degen, E. A., Heisey, K. L., & Garnett, R. "Conjoint psychometric field estimation for bilateral audiometry." *Behavior Research Methods*, 51 (2019), 1271–1285. DOI: 10.3758/s13428-018-1062-3.
- Barbour, D. L., Howard, R. T., Song, X. D., Metzger, N., **Sukesan, K. A.**, et al. "Online machine learning audiometry." *Ear and Hearing*, 40(4) (2019), 918–926. DOI: 10.1097/AUD.0000000000000669.